SDG BLOCKCHAIN ACCELERATOR

Mentor Feedback Report : TradeChainX

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Program: EMURGO : UNDP Blockchain Accelerator (Cohort 2)
Project: TradeChainX
Focus Area: Carbon Markets, MRV, Tokenized Commodities, SME Finance
Mentorship Focus: MRV System Design, Lifecycle Integrity, Market Readiness
Total Mentoring Sessions: 5
Mentor: Ann Lu

1. Where the Project Started

TradeChainX entered Cohort 2 with a broad and ambitious vision: creating a multi-asset platform that connects smallholders and agri-SMEs to carbon markets and commodity finance. From the start, the team understood that *access* was the problem , access to finance, access to markets, access to credibility.

What was less clearly defined at the beginning was what TradeChainX should be first.

Early conversations revealed three competing identities:

- A carbon credit marketplace
- A commodity trading platform
- A DeFi-style financing layer for SMEs

Each of these directions was valid, but trying to build all of them at once created architectural and conceptual risk. The mentoring work therefore focused on narrowing the scope to something concrete, defensible, and testable: Measurement, Reporting, and Verification (MRV) as the foundation layer.

2. Mentoring Focus Shift: From “Platform” to “Lifecycle”

Rather than treating MRV as a feature, mentoring reframed it as the *core system*.

The key reframing question introduced early was:

> “What does the system need to guarantee, even if the marketplace or UI disappears?”

This question helped shift discussions away from surface-level features and toward lifecycle guarantees:

- When is a project considered valid?
- Who is allowed to verify it?

- Under what conditions can credits be minted?
- How do credits definitively leave circulation?

This reframing shaped the rest of the mentoring process.

3. Mentoring Sessions and Key Turning Points

Session 1 : Defining the MRV Boundary

The first session focused on drawing a hard boundary around MRV.

We walked through the full carbon credit lifecycle as it exists off-chain today and mapped it onto what *must* be enforced on-chain versus what can remain procedural or informational.

At this stage, the team was still thinking in terms of “carbon credit tokens.” The mentor feedback pushed them to instead think in terms of state transitions:

- A project is registered
- A project is verified
- A project becomes eligible to mint
- Credits are minted
- Credits are retired

The important insight from this session was that minting is not the start of the lifecycle , it is the end of verification. This led to the decision to model verification explicitly, rather than implicitly through backend logic.

Session 2 : Smart Contract Separation and State Modeling

The second session went deep into how to represent lifecycle states safely using Cardano’s UTxO model.

Initially, the team had explored a single-contract approach. While simpler, this made reasoning about permissions and transitions difficult. Mentoring feedback encouraged separation by responsibility:

- One validator for project registration
- One validator for verification actions
- A distinct minting policy tied to verified state

This session involved careful discussion of trade-offs. Multiple validators increase complexity, but they also:

- Make audits easier
- Reduce the blast radius of mistakes

- Allow clearer reasoning about who can do what

The team agreed that clarity and safety were more important than minimalism, especially in a domain where trust and credibility are paramount.

Session 3 : Evidence, Privacy, and “What Not to Put on Chain”

The third session focused less on contracts and more on data.

TradeChainX deals with evidence: MRV reports, measurements, documents, and third-party attestations. A natural temptation was to push as much of this as possible on-chain to increase transparency.

Mentoring deliberately pushed back on this instinct.

The discussion centered on three principles:

1. **Integrity over visibility** : prove data hasn’t changed, not that everyone can see it.
2. **Auditability over immutability** : auditors need traceability, not raw blobs.
3. **Privacy by default** : especially when working with smallholders and SMEs.

This led to a clear design direction:

- Evidence remains off-chain
- Cryptographic hashes anchor evidence on-chain
- On-chain state only reflects what is necessary to enforce lifecycle rules

This session significantly reduced future risk around data protection, scalability, and regulatory compliance.

Session 4 : Retirement, Double Counting, and Market Credibility

By the fourth session, the team had a solid MRV core, but one issue remained central: **double counting**.

The discussion here was blunt. If credits can be reused, re-sold, or ambiguously retired, the entire system loses credibility , regardless of how good the verification process is.

Mentoring therefore focused on retirement semantics:

- What does it mean for a credit to be “used”?
- How do we prevent accidental reuse?
- How do we prove to an external party that a credit is no longer valid?

The outcome of this session was a clear commitment to explicit retirement logic, enforced at the contract level, not as a marketplace feature. Retirement became a first-class lifecycle event, not an afterthought.

Session 5 : Market Readiness and Sequencing

The final session stepped back from the technical details and looked at the bigger picture.

Rather than asking “what else can we build?”, the focus was:

- What can realistically be piloted?
- Who needs to trust this system first?
- What evidence would convince a skeptical buyer or institution?

Mentoring guidance emphasized sequencing:

- Start with MRV credibility
- Demonstrate lifecycle enforcement
- Only then layer on trading, financing, or DeFi components

The team aligned on using the MRV PoC as a *credibility anchor* for future partnerships, rather than rushing toward a full marketplace prematurely.

4. Overall Mentor Assessment

TradeChainX showed strong adaptability during Cohort 2. The most notable progress was not in the volume of features delivered, but in the **clarity of system boundaries** and the discipline around lifecycle enforcement.

Key strengths observed:

- Willingness to narrow scope to get fundamentals right
- Strong engagement with architectural trade-offs
- Growing understanding of how credibility is built in carbon markets

Key risks to watch going forward:

- Pressure to expand scope too quickly
- Overloading the system with features before trust is established
- Communicating technical guarantees clearly to non-technical stakeholders

From a mentor perspective, TradeChainX made the right decision to anchor its platform in MRV first. This gives the project a solid foundation on which more complex market and financing layers can later be built without undermining trust.

5. Mentor Conclusion

TradeChainX is best understood not as a marketplace, but as an **infrastructure project**. During Cohort 2, mentoring helped the team shift from a broad platform vision to a focused, defensible system that prioritizes lifecycle integrity and auditability.

If the team maintains this discipline, TradeChainX is well positioned to engage with institutional partners, carbon market stakeholders, and future pilots that require more than surface-level blockchain integration.